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Department of Health and Human Services
Office of the National Coordinator for Health Information Technology
Attention: Request for Information: HHS Health Sector AI RFI
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Re: Docket: HHS Health Sector AI RFI, Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care

Dear Assistant Secretary Keane:

Eli Lilly and Company (Lilly) appreciates the opportunity to inform HHS's approach to accelerating AI adoption in clinical care. As a leading biopharmaceutical innovator with extensive AI deployment across drug discovery, clinical trials, manufacturing quality, and patient support, Lilly offers practical insights on regulation, reimbursement, and research & development policy.

Lilly's AI Leadership

Lilly is leveraging AI throughout the drug lifecycle through strategic partnerships with NVIDIA (AI-accelerated drug discovery infrastructure),¹ Isomorphic Labs (drug discovery), OpenAI (antimicrobial discovery), BigHat Biosciences (antibody design), and Thrive Global (patient behavior change). Internally, we have deployed AI for manufacturing quality, clinical trial optimization, and post-market surveillance. This operational experience informs our recommendations for accelerating AI adoption while safeguarding public health, privacy, and trust.

Three Pillars for AI Acceleration

Pillar 1: Regulatory Clarity Enables Innovation

Lilly supports risk-proportionate, adaptive regulatory frameworks that provide certainty while maintaining rigorous safety and efficacy standards. The Food and Drug Administration (FDA) should also leverage AI internally to streamline review processes while maintaining transparent governance and protecting confidential information.

Pillar 2: Reimbursement Modernization Is Critical

Legacy fee-for-service (FFS) payment systems represent the single largest barrier to AI adoption in clinical care. The fundamental problem: FFS systems are structurally misaligned with AI's value proposition. AI tools deliver value through preventive interventions that avoid future costs, efficiency gains that reduce unnecessary procedures, and quality improvements through earlier detection, none of which generate billable services under FFS economics, which reimburses based on volume rather than outcomes.² Without payment reform, AI innovations may remain throttled by reimbursement inertia regardless of clinical value or regulatory approval.

Pillar 3: Strategic Research & Development (R&D) Investment Accelerates Translation

Public-private partnerships and targeted research funding can bridge the gap between AI development and real-world clinical implementation.

¹ See Lilly and NVIDIA Announce Partnership to Accelerate AI-Driven Drug Discovery (Nov. 2023), available at <https://investor.lilly.com/news-releases/news-release-details/lilly-and-nvidia-announce-partnership-accelerate-ai-driven-drug>

² HHS Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care, 90 Fed. Reg. 60108, 60109 (Dec. 23, 2025) (acknowledging that AI risks being "throttled by payment inertia, not science or regulation").

Priority Recommendations

I. REGULATORY CLARITY

1. Expand Adaptive Clinical Trial Frameworks

Current Barrier: Traditional clinical trial regulations (e.g., 21 CFR Parts 312, 314) assume static protocols, fixed endpoints, and predetermined sample sizes, creating friction for AI-enabled adaptive designs that optimize recruitment in real time.

FDA should clarify acceptable approaches to AI-driven adaptive randomization, dynamic endpoint modification, and continuous learning algorithms, including AI-driven dosing recommendations such as precise timing of next dose administration. This clarity would accelerate precision prevention therapies requiring large, long-duration studies.

2. Modernize HIPAA De-identification Standards for AI Era

Current Barrier: The Health Insurance Portability and Accountability Act (HIPAA) de-identification framework under 45 CFR §164.514 is structurally incompatible with modern data operations. The safe harbor's 18-identifier checklist neither accounts for AI-enabled re-identification risks nor provides flexibility for privacy-preserving techniques that may offer stronger protections. More critically, the expert determination pathway requires individualized expert certification for every dataset configuration, which is an operational impossibility for organizations running continuous, enterprise-scale tokenization and pseudonymization pipelines.

HHS should modernize §164.514 in three targeted ways: (1) establish a formal pseudonymization tier that provides regulatory recognition for tokenized data held under controlled re-linkage architectures, rather than treating it identically to unprotected personal health information (PHI); (2) replace per-instance expert determinations with a certification framework developed by the National Institute of Standards and Technology (NIST) for approved privacy-preserving methodologies (including differential privacy, federated learning, and synthetic data generation) that satisfy the "very small risk" standard by design; and (3) issue affirmative guidance on permissible use of health data for AI model training, closing the significant interpretive gap that currently drives inconsistent compliance practices and impedes beneficial AI development. These changes would reduce compliance friction while providing more consistent and technically grounded privacy protections than the existing framework achieves.

3. Establish Platform Technology Designations for AI-Enabled Therapeutics

Current Barrier: AI platforms that generate multiple therapeutic candidates must undergo separate, redundant regulatory review for each candidate, despite using the same validated underlying technology.

FDA should expand its Platform Technology Designation program to AI-enabled drug discovery platforms, allowing one-time validation of core AI methodology with streamlined review for subsequent therapeutics generated using the validated platform. This accelerates development of AI-discovered therapeutics, particularly for rare diseases and areas of high unmet need.

4. Clarify Governance for Foundation Models and Third-Party AI

Current Barrier: Many pharmaceutical AI applications leverage foundation models (e.g., AlphaFold or large language models) developed by third parties, but current guidance doesn't address validation requirements when sponsors lack full access to training data or architecture.³

³ PhRMA, Use of AI in the Drug Lifecycle and Regulatory Activities at 3 (2025) ("Many AI tools, including foundation models, used in drug development are created by third parties, which can present challenges and opportunities.").

FDA should issue guidance addressing third-party AI tools and foundation models (like those Lilly uses in partnership with OpenAI and Isomorphic Labs), establishing tiered validation approaches and model registries to enable efficient oversight without requiring full architectural transparency from technology partners. This will make it easier for companies developing innovative medicines to partner with companies developing innovative AI models, with each leveraging their core strengths to benefit patients. For example, Lilly's collaboration with OpenAI uses foundation models to discover novel antimicrobials; these models were trained on datasets outside our control, but performance-based validation would enable oversight without requiring Lilly to develop foundation models itself.⁴

5. Provide Additional Examples of Credibility Assessment Framework Application

Current Barrier: FDA's credibility assessment framework provides a strong foundation, but sponsors need additional examples showing application to diverse use cases to ensure consistent implementation.

FDA's draft guidance on AI to support regulatory decision-making provides a strong foundation. Additional examples showing application to diverse use cases (real-world evidence, exploratory endpoints, combination products, biologics manufacturing) would accelerate industry adoption and regulatory consistency. This reduces regulatory uncertainty and accelerates market entry for validated tools.

II. REIMBURSEMENT MODERNIZATION

6. Create Medicare Coverage Pathways for AI-Enabled Preventive Diagnostics

Current Barrier: Medicare's statutory framework (42 USC §1395y(a)(1)(A)) has been interpreted to exclude coverage for "screening tests," creating a categorical barrier to AI-enabled diagnostics that identify disease risk before symptoms manifest.⁵

The Centers for Medicare & Medicaid Services (CMS) should clarify its coverage authority to distinguish AI-enabled diagnostic procedures (targeted testing based on validated biomarkers, genetic risk, or AI-predicted risk scores) from traditional screening tests (population-wide testing without risk stratification). Example: AI-powered digital cognitive assessments can detect early cognitive decline and enable care planning yet lack clear Medicare coverage pathways because they may be classified as screening rather than diagnostic tools.⁶ This reform would enable earlier intervention rather than reactive late-stage treatment.

7. Expand Voluntary Value-Based Payment Models for AI Clinical Decision Support

Current Barrier: Healthcare systems have no direct financial incentive to adopt AI clinical decision support tools under FFS payment; efficiency gains from AI may actually reduce revenue (fewer hospital days, fewer procedures). While CMS's WISer model tests AI-enabled prior authorization to reduce waste, no parallel incentives exist for AI tools that proactively improve clinical outcomes.⁷

⁴ See Lilly, Lilly collaborates with OpenAI to discover novel medicines to treat drug-resistant bacteria (Apr. 2024), available at <https://investor.lilly.com/news-releases/news-release-details/lilly-collaborates-openai-discover-novel-medicines-treat-drug>.

⁵ 42 U.S.C. § 1395y(a)(1)(A). CMS has historically interpreted this provision to exclude coverage for screening tests in asymptomatic individuals unless Congress has specifically authorized such screening, as it has for mammography (42 U.S.C. § 1395x(jj)), colorectal cancer screening (42 U.S.C. § 1395x(pp)), and certain other preventive services.

⁶ Digital cognitive assessments (DCAs) leverage AI to standardize cognitive testing, streamline administration and interpretation, and enable better disease tracking and care planning. Medicare requires cognitive assessments during Annual Wellness Visits, yet validated digital tools face reimbursement uncertainty despite their potential to improve detection of cognitive decline, particularly as primary care providers increasingly perform assessments beyond specialist settings. See Coalition to Evaluate Older Adults Innovations (CEOi), Digital Cognitive Assessments, available at <https://alzdiagnostichub.org/digital-cognitive-assessments/>; see also Linus Health (example of AI-powered DCA platform for early cognitive impairment detection).

⁷ See CMS, Wasteful and Inappropriate Service Reduction (WISer) Model (Dec. 2025), available at <https://www.cms.gov/priorities/innovation/innovation-models/wiser> (testing AI-enabled prior authorization to reduce inappropriate utilization); see also Lilly, Aspen Ideas: Health—Imagine a World Without Disease, Fireside Chat Talking Points at 7-8 (June 2025) (advocating for CMS to use Medicare Advantage risk-adjustment factors and MIPS to incentivize preventive interventions).

CMS should create complementary alternative payment models (via 42 CFR Part 425, §414.1380) that reward healthcare systems for adopting validated AI tools that demonstrably improve outcomes or reduce total cost of care. Medicare Advantage risk-adjustment bonuses, Merit-based Incentive Payment System (MIPS) quality points, and shared savings programs should explicitly incentivize evidence-based AI adoption that enhances care quality, not just reduces utilization. Balanced incentives would be created for AI adoption focused on both waste reduction and outcome improvement.

8. Extend Medicare Eligibility for Conditions Requiring Early Intervention

Current Barrier: Medicare eligibility begins at age 65 (with exceptions for end-stage renal disease and amyotrophic lateral sclerosis (ALS)), creating barriers to early intervention for chronic diseases with long preclinical phases. The recently introduced BRIDGE Act addresses Alzheimer's disease by eliminating waiting periods for symptomatic young-onset patients, but broader payment system reforms are needed as AI technologies increasingly enable identification of high-risk asymptomatic individuals requiring preventive intervention decades before age 65.⁸

Congress should pass the BRIDGE Act and use it as a model for condition-specific Medicare eligibility expansions as scientific evidence supports early intervention. As AI-powered precision medicine advances, HHS should establish frameworks ensuring payment systems can accommodate validated preventive therapies for high-risk populations identified through genetic testing, biomarker analysis, and AI-enabled risk prediction, enabling intervention during optimal treatment windows regardless of age-based eligibility thresholds—an approach that improves patient outcomes and reduces long-term program costs.

9. Prevent Misuse of AI in Coverage Determination

Current Barrier: Growing concern that payers may deploy AI tools to automate coverage denials or accelerate prior authorization rejections without adequate clinical oversight or transparency.⁹

HHS should establish guardrails for payer use of AI in coverage decisions, including transparency requirements (disclosure of AI system purpose, data sources, decision criteria), clinical oversight mandates (meaningful physician review, with AI authorized to approve coverage but all denials requiring human clinician determination), timely appeals processes, and bias auditing. AI should improve care delivery, not automate barriers to access through coverage denial.¹⁰ Public trust is built by preventing algorithmic barriers to care while allowing AI to reduce administrative burden for routine approvals.

III. RESEARCH & DEVELOPMENT INVESTMENT

10. Accelerate Surrogate Endpoint Validation for Preventive Therapies

Current Barrier: Preventive therapies often require large, long-duration trials (5-10 years with thousands of participants), making development timelines incompatible with rapid AI innovation cycles.¹¹

The National Institutes of Health (NIH) and FDA should fund research developing and validating AI-discovered biomarkers and digital phenotypes that can serve as surrogate endpoints for preventive therapies. Apply the oncology paradigm (where 70% of cancer drug approvals use surrogate endpoints like

⁸ See Bridging Relief in Delayed Government Enrollment for Young-Onset Alzheimer's Disease Act of 2025 (BRIDGE Act), 119th Cong. (2025) (proposing to eliminate disability insurance and Medicare waiting periods for individuals with young-onset Alzheimer's, following the model of existing exceptions for amyotrophic lateral sclerosis). Lilly supports this important legislation. More broadly, as AI technologies enable increasingly sophisticated risk stratification and earlier disease detection across multiple therapeutic areas, payment systems must evolve to enable coverage for validated preventive interventions when scientific evidence supports early treatment of high-risk asymptomatic populations.

⁹ PhRMA, Artificial Intelligence Policy Priorities at 1 (Aug. 2025) ("AI-enabled solutions should be designed to help patients, providers, and purchasers make informed decisions about the tools themselves and their health care. Such AI tools should be used to improve patient outcomes—not to automate or accelerate barriers to care through coverage denial nor reinforce flawed utilization management practices.").

¹⁰ *IBID*

¹¹ See Lilly, Aspen Ideas: Health—Imagine a World Without Disease, Fireside Chat (June 2025) (explaining that clinical trials for primary and secondary prevention "often require more people and more time").

response rate or progression-free survival, reducing approval timelines by 1-1.5 years) to other disease areas including neurodegenerative diseases, cardiovascular disease, and metabolic disorders.¹² This would dramatically reduce time and cost to validate preventive therapies, making development economically viable.

11. Invest in Bias Detection and Mitigation Research

Current Barrier: AI algorithms trained on historical data may perpetuate or amplify existing health disparities; detecting and mitigating biases require specialized research expertise.

NIH should establish a dedicated research program focused on fairness metrics for clinical AI, algorithmic auditing methods, bias mitigation techniques, and prospective studies measuring real-world impact of AI tools on health disparities. AI algorithms used in drug development and clinical care must be free from bias and avoid exacerbating health inequities. For example, in Alzheimer's disease, which disproportionately impacts women and certain minority populations, AI algorithms must account for sex-based and racial differences in disease presentation, progression, and treatment response to ensure equitable outcomes.¹³ Ensuring AI tools reduce rather than exacerbate health disparities is essential to building public trust.

12. Develop Real-World Performance Monitoring Infrastructure

Current Barrier: AI tools that perform well in controlled research settings may degrade when deployed in diverse real-world environments; robust post-deployment monitoring remains methodologically challenging.

HHS should fund research on automated performance monitoring systems, benchmarking frameworks, and rapid validation protocols for algorithm updates. Cooperative agreements with health systems, academic medical centers, and AI developers can create shared infrastructure for ongoing performance evaluation. This enables continuous improvement while maintaining safety and building payer confidence in AI reliability.

13. Foster International Harmonization

Current Barrier: Divergent international regulatory approaches to AI create duplicative requirements and slow global development timelines.

HHS should prioritize engagement through the International Council for Harmonisation (ICH), International Coalition of Medicines Regulatory Authorities (ICMRA), International Medical Device Regulators Forum (IMDRF), and Council for International Organizations of Medical Sciences (CIOMS) to align on common AI terminology, harmonized validation standards, and mutual recognition agreements. Recent FDA collaborations with Health Canada and the United Kingdom's Medicines and Healthcare products Regulatory Agency (UK MHRA) on machine learning-enabled medical devices demonstrate the value of this approach.¹⁴ Regulatory burden is reduced for developers and accelerates global access to AI innovations.

¹² Id. at 5 ("About 70% of all FDA approvals for cancer medicines come via trials that measure surrogate endpoints (RR and PFS) rather than overall survival. As a result, using those surrogates meant those approvals came 1-1.5 years sooner than if they had used overall survival—critical time savings for a fatal disease."); citing Hwang TJ, et al., Association Between FDA and EMA Expedited Approval Programs and Therapeutic Value of New Medicines: Retrospective Cohort Study, 366 BMJ 14340 (2019), available at <https://pmc.ncbi.nlm.nih.gov/articles/PMC6503556/>.

¹³ Alzheimer's disease affects nearly two-thirds of Americans living with the disease are women, and Black Americans are about twice as likely to develop Alzheimer's compared to white Americans. See Alzheimer's Association, 2024 Alzheimer's Disease Facts and Figures, 20 Alzheimer's & Dementia 3708 (2024). AI algorithms used in Alzheimer's drug development and clinical decision support must account for these disparities to ensure equitable performance across demographic groups.

¹⁴ See FDA, Health Canada, and UK MHRA, Good Machine Learning Practice for Medical Device Development: Guiding Principles (Oct. 2021), available at <https://www.fda.gov/medical-devices/software-medical-device-samd/good-machine-learning-practice-medical-device-development-guiding-principles>; FDA, Proposed Regulatory Framework for Modifications to Artificial Intelligence/Machine Learning (AI/ML)-Based Software as a Medical Device (SaMD), Discussion Paper and Request for Feedback (Apr. 2019), available at <https://www.fda.gov/media/122535/download> (discussing predetermined change control plans).

14. Support Responsible FDA Adoption of AI for Review Efficiency

Current Barrier: As FDA scales AI adoption, lack of transparent governance creates uncertainty about protecting confidential information, ensuring data quality, and maintaining human oversight.

FDA should continue deploying AI tools to streamline workflows and improve review consistency, while establishing transparent governance frameworks addressing six critical considerations:

- (1) Protect confidential commercial information, ensure AI tools are never trained on sponsor data without permission, and ensure third-party vendors cannot access proprietary information;¹⁵
- (2) Maintain accountability, publish annual transparency reports on which AI tools are deployed, how they contribute to decisions, and validation metrics;¹⁶
- (3) Ensure data quality, establish standards for training data representativeness and conduct bias audits;¹⁷
- (4) Require human oversight, mandate human review for novel therapeutics, expedited pathways, and safety signals;¹⁸
- (5) Achieve cross-center consistency, create agency-wide AI governance body to harmonize deployment;¹⁹ and
- (6) Secure systems, protect against cyberattacks and adversarial manipulation.²⁰

Regular stakeholder engagement on FDA's AI use is essential to building trust and sharing best practices.²¹ It enables FDA to realize AI efficiency gains while maintaining gold standard review and protecting sponsor confidential information.

Responses to Key HHS Questions

Question 1: What are the biggest barriers to AI adoption in clinical care?

Based on Lilly's experience: (1) Reimbursement inertia (FFS payment systems fail to reward AI-enabled prevention, efficiency gains, or quality improvements), (2) Regulatory uncertainty (unclear validation standards), (3) Data access constraints (privacy regulations and fragmentation), (4) Liability ambiguity (unclear responsibility allocation), and (5) Workforce readiness (insufficient AI literacy). Reimbursement is the dominant barrier; many healthcare systems identify valuable AI tools but cannot justify adoption under current payment models.

Question 3: What novel legal issues exist for non-medical device AI?

Two critical issues: (1) intellectual property (IP) protection during regulatory review, current trade secret protections may be inadequate for AI algorithmic submissions to FDA; (2) Federated learning governance, unclear rules on model ownership, institutional contribution recognition, and consent requirements when data never leaves originating institutions. HHS should convene working groups to develop model liability

¹⁵ PhRMA, Use of AI in the Drug Lifecycle and Regulatory Activities at 4 (2025) ("FDA must ensure AI tools deployed within the Agency are not trained on data submitted by industry or used for any other purpose without express permission of the sponsor. Furthermore, FDA must ensure compliance with all applicable laws to protect sponsors' data and prevent unauthorized access or use, including third-party access.").

¹⁶ . at 4-5 ("It is important for FDA to share experiences regarding how AI models used by the Agency are being trained and developed, what criteria the Agency uses to assess AI model credibility and trustworthiness, what specific AI models it uses (including from third-party developers), how the AI model contributes to regulatory activities, and how FDA uses human-in-the-loop to ensure appropriate oversight and accuracy of AI models.").

¹⁷ Id. at 5 ("Data quality is paramount for training AI models, and FDA must ensure the data is fit-for-purpose.").

¹⁸ Id. at 5 ("FDA should continue to employ a reasonable human-in-the-loop process with proper training and accountability for clear and effective oversight when using AI tools to assist with regulatory activities.").

¹⁹ Id. at 5-6 ("FDA should ensure that AI tools are consistently applied across the Agency, as appropriate, so that the benefits these tools offer can be fully realized. Ensuring consistency in implementation, particularly when products require review by more than one center (e.g., combination products), will be essential for achieving Agency-wide benefits. Siloed use of AI could lead to divergent results and delayed implementation.").

²⁰ Id. at 6 ("AI systems utilized by FDA must be secure. FDA should ensure that the AI systems are not penetrable or susceptible to outside influence, which could impact outputs or affect the performance of the trained models and processes.").

²¹ Id. at 5 ("There should continue to be an opportunity for regular stakeholder engagement around FDA's adoption and use of AI. Furthermore, FDA should continue to seek stakeholder input and provide opportunities for direct communication on FDA's use of AI, including how the Agency will address concerns and provide clarity on the use and limitations of AI in regulatory activities. Transparent communication will be key to maintaining scientific integrity and public trust.").

frameworks, strengthen confidentiality protections for AI submissions, and issue guidance clarifying HIPAA applicability to federated learning.

Question 5: How can HHS best support private sector AI innovation?

HHS can support innovation by demonstrating responsible AI adoption within its own operations. When FDA successfully deploys AI tools, it sets standards, builds confidence, creates shared learning, and identifies common barriers. However, this requires transparent communication through stakeholder engagement.²² FDA should publish case studies, host workshops, maintain open dialogue, and work with NIST and international regulators on best practices. See Recommendation #14.

Question 7: What influences AI adoption decisions within healthcare organizations?

Key decision-makers include C-suite leadership (strategic AI investment decisions based on return on investment (ROI)), clinical department chairs (specialty-specific adoption), information technology (IT) governance committees (technical integration), value analysis committees (cost-effectiveness), and front-line clinicians (ultimate determinants through usage). Critical insight: Even when leadership approves AI adoption, implementation fails without front-line clinician buy-in. Primary administrative hurdles include IT integration complexity (legacy electronic health record (EHR) systems lack standardized application programming interfaces (APIs)), credentialing processes (no clear approach for “credentialing” AI tools), change management resources (limited capacity for training and workflow redesign), and contracting delays (procurement processes poorly suited to AI tools requiring ongoing updates).

The Stakes Are High: If HHS acts decisively, the United States will maintain global leadership in healthcare AI with enormous benefits for patients. If payment inertia and regulatory uncertainty persist, innovation will migrate offshore and American patients will lose access to transformative technologies developed elsewhere. Lilly stands ready to partner with HHS in building an AI-enabled healthcare system that is innovative, equitable, and worthy of public trust.

Sincerely,



Diogo Rau
Executive Vice President and Chief Information and Digital Officer
Eli Lilly and Company

²² Id. at 5.